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Indian Standard

**DETERMINATION OF
FAT BY THE GERBER METHOD**

PART II MILK PRODUCTS

(First Revision)

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**BUREAU OF INDIAN STANDARDS
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**DETERMINATION OF
FAT BY THE GERBER METHOD**
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(First Revision)

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Indian Standard
DETERMINATION OF
FAT BY THE GERBER METHOD
PART II MILK PRODUCTS
(*First Revision*)

0. FOREWORD

0.1 This Indian Standard (Part II) (First Revision) was adopted by the Indian Standards Institution on 30 May 1977, after the draft finalized by the Dairy Products Sectional Committee had been approved by the Agricultural and Food Products Division Council.

0.2 The Gerber method for determination of fat in milk and milk products depends on the liberation of the fat by the action of sulphuric acid on milk or milk products in specially shaped, calibrated glass containers, called butyrometers which are then centrifuged to aid the separation of fat, the volume of which is finally read off against the percentage scale etched on the butyrometer. This method is widely used in India for the rapid determination of fat in milk and other milk products when a large number of samples is to be analyzed at a time.

0.2.1 With the increasing application of the Gerber method for determination of fat in milk and milk products for commercial need in India, it has become necessary to prescribe a uniform method, using standard apparatus designed to meet the Indian conditions, so that the results obtained in different laboratories and by different workers are comparable within a reasonable degree of accuracy.

0.3 This standard was first issued in 1958 as IS : 1224-1958*. It has now been decided to issue the revised version of the standard into two parts. Part I deals with determination of fat in milk. This part (Part II) deals with the determination of milk fat in milk products, such as cream, milk powder and cheese, the last two products had not been included in the earlier version of the standard.

0.4 It should be emphasized that the methods prescribed in this standard are designed for routine purposes only, and while these give results comparable with gravimetric methods, these cannot be substituted for gravimetric methods, which will have to be used for reference purposes.

*Determination of fat in milk, evaporated (unsweetened) milk, separated milk skimmed milk, buttermilk and cream by the Gerber method.

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0.5 In the formulation of this standard, considerable assistance has been derived from BS 696 : Part II : 1969 Gerber method for the determination of fat in milk and milk products : Part II Methods, issued by the British Standards Institution.

0.6 In reporting the result of a test or analysis made in accordance with this standard, if the final value, observed or calculated, is to be rounded off, it shall be done in accordance with IS : 2-1960*.

1. SCOPE

1.1 This standard (Part II) prescribes the procedure for the determination of fat in milk products by the Gerber method.

1.1.1 The methods for milk powders is applicable to powders containing not less than 10 percent fat.

2. APPARATUS

2.1 The apparatus given in **2.1.1** to **2.1.8** conforming to the provisions of IS : 1223 (Part I)-1970†, IS : 1223 (Part II)-1972‡ and IS : 1223 (Part III)-1977§ are required.

2.1.1 *Butyrometer, 70 Percent Scale* — for estimating fat in cream.

2.1.2 *Butyrometer, 6 Percent, 8 Percent and 10 Percent Scale* — for estimating fat in milk powders.

2.1.3 *Cheese Butyrometer, 40 Percent Scale* — for estimating fat in cheese.

2.1.4 *10-ml Pipette or Automatic Measure for Sulphuric Acid*

2.1.5 *1-ml Pipette or Automatic Measure for Amyl Alcohol*

2.1.6 *Stoppers for Butyrometers*

2.1.7 *Centrifuge*

2.1.8 *Water-Bath*

2.2 Other Apparatus

2.2.1 *Weighing Balance* — suitable for weighing to 0.01 g.

2.2.2 *Small Scoop* — of suitable material (for example, aluminium, nickel or plastics). It is convenient if this scoop has a counterpoise.

*Rules for rounding off numerical values (revised).

†Specification for apparatus for determination of milk fat by Gerber method : Part I Butyrometers and stoppers (first revision).

‡Specification for apparatus for determination of milk fat by Gerber method : Part II Pipettes and automatic measures (first revision).

§Specification for apparatus for determination of milk fat by Gerber method : Part III Centrifuges and water-baths (first revision).

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2.2.3 Stemless Funnel

2.2.4 Glass Rod — of such diameter (6 mm is convenient) that the rod will pass through the hole at the base of the funnel (**2.2.3**).

2.2.5 Small Camel Hair Brush

2.2.6 Wash Bottles — containing cold and hot distilled water (*see* IS : 1070-1977*) at about 30 to 40°C and 70°C.

2.2.7 Stoppered Funnel

2.2.8 Grater or Pestle and Mortar

3. REAGENTS

3.1 Sulphuric Acid — Sulphuric acid shall have a density of 1.807 to 1.812 g/ml at 27°C corresponding with a concentration of sulphuric acid from 90 to 91 percent by mass.

3.1.1 The sulphuric acid shall be colourless or not darker than pale amber in colour.

3.1.2 When diluted with distilled water to a density of 1.4 g/ml, not more than a very slight turbidity shall occur.

3.2 Amyl Alcohol — It shall conform to Grade 1 of IS : 360-1964†.

4. METHOD FOR CREAM

4.1 Mixing of Sample — Stir the sample thoroughly but not so vigorously as to cause undue froth or churning. If the cream is very thick, warm to a temperature between 30°C and 40°C to facilitate mixing. Mix immediately before weighing the required amount of cream for the test.

4.1.1 If it is not possible to achieve complete mixing, the sample should not be tested by this method.

4.2 Method 1

4.2.1 Weighing of Sample — Weigh 5.00 ± 0.01 g of sample into a dried and tared beaker of about 50-ml capacity.

4.2.2 Addition of Acid and Alcohol — Add in small quantities freshly prepared mixture of one volume of sulphuric acid (*see* **3.1**) and one volume of distilled water to the cream in the beaker. Dissolve the cream and transfer the mixture carefully, with the help of a dry funnel, to the butyrometer for testing cream (*see* **2.1.1**). Rinse the beaker about six times

*Specification for water for general laboratory use (*second revision*).

†Specification for amyl alcohol (*revised*).

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with small quantities of the dilute acid to make sure that all cream has been transferred to the butyrometer. Altogether 18 to 20 ml of the diluted acid should suffice to fill the butyrometer, leaving sufficient space for the addition of amyl alcohol and mixing. Add one millilitre of amyl alcohol (see 3.2) by means of the 1-ml pipette or the automatic measure (see 2.1.5). Do not wet the neck of the butyrometer with alcohol.

4.2.3 Insertion of Stopper — Close the neck of the butyrometer firmly with the stopper (see 2.1.6) without disturbing the contents. When a double-ended stopper is used, screw it in until the widest part is at least level with the top of the neck. When a lock stopper is used, insert it until the rim is in contact with the neck of the butyrometer.

4.2.4 Mixing of Contents — Shake the butyrometer carefully, without inverting it, until the contents are thoroughly mixed, the curd is dissolved and no white particles are seen in the liquid. Then invert the butyrometer a few times to mix the contents thoroughly.

NOTE — When a large number of samples is to be mixed, shake the butyrometers in a protected stand until the contents are thoroughly mixed and no white particles are seen. Invert once or twice during the process.

4.2.5 Temperature Adjustment — Transfer the butyrometer quickly, with the bulb uppermost, into a water-bath having a temperature of $65 \pm 2^{\circ}\text{C}$ and leave it there for not less than 3 minutes and not more than 10 minutes. Take care to have the water level in the bath above the top of the fat column in the butyrometer. Meanwhile, adjust the stopper so that the fat column should be on the scale after centrifuging.

4.2.6 Centrifuging — Take the butyrometer out of the water, dry it with a cloth and transfer it to the centrifuge, placing two butyrometers diametrically opposite so as to balance the rotating disc. Centrifuge at the maximum speed for 5 minutes. Bring the centrifuge to stop gradually. Transfer the butyrometers, stoppers downwards, into a water-bath having a temperature of $65 \pm 2^{\circ}\text{C}$ as in 4.2.5, and allow the butyrometers to stand in the water-bath for at least 3 minutes and not more than 10 minutes.

4.2.7 Reading of Butyrometer — Before taking a reading, adjust the position of the fat column to bring the lower end of the column on to a main graduation mark. When double-ended stoppers are used, do this by slightly withdrawing the stopper and not by forcing it further into the neck. Note the scale readings corresponding to the lowest point of the fat meniscus and the surface of separation of the fat and acid; the difference between the two readings gives the percentage by mass of fat in the cream. When readings are being taken, hold the butyrometer with the graduated portion vertical, keep the point read in level with the eye, and then read the butyrometer to the nearest 0.5 percent that is, half of the smallest scale division.

4.2.8 Second Centrifuging — If after centrifuging there is not a sharp dividing line between the fat and the acid, or if the acid layer is not clear,

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repeat the temperature adjustment (*see 4.2.5*) and centrifuging (*see 4.2.6*) before taking the reading.

4.2.9 Precautions — If a fluffy layer is observed at the base of the fat column in the butyrometer, reject the test. Examine the stopper to see if it is in good condition, repeat the test and take care to ensure that the curd is completely dissolved.

4.2.9.1 If the fat column is so dark as to make reading difficult, reject the test and check the strength of the sulphuric acid.

4.2.9.2 If a large number of samples has to be tested, it is preferable to use automatic measures for measuring the sulphuric acid and amyl alcohol, especially the latter, otherwise there is a possibility of injurious effects to the health arising from the inhalation of amyl alcohol vapours by the use of 1-ml pipette.

4.2.10 Check Reading — Replace the butyrometer in the bath for another 3 minutes and then take a check reading of the butyrometer as rapidly as possible.

4.3 Method 2

4.3.1 Addition of Acid to Butyrometer — Transfer 10 ml of sulphuric acid (*see 3.1*) into the butyrometer by means of the 10-ml pipette for sulphuric acid or the automatic measure (*see 2.1.4*) taking care not to wet the neck of the butyrometer with the sulphuric acid.

4.3.2 Weighing of Sample — Mix, and immediately weigh 5 ± 0.01 g of sample into the butyrometer without soiling the neck, using any suitable form of support for the butyrometer on the balance.

4.3.3 Addition of Water and Amyl Alcohol — Add about 6 ml of the hot water (70°C) to the butyrometer. Measure 1 ml of amyl alcohol into the butyrometer by means of the alcohol pipette or automatic measure. Adjust the level of the contents to about 5 mm below the shoulder by further additions of hot water. In no circumstances shall the amyl alcohol be added to the butyrometer before the cream.

4.3.4 Follow the procedures described in **4.2.3** to **4.2.10**.

5. METHOD FOR MILK POWDERS

5.1 Weighing of Sample — Mix the sample thoroughly, and weigh 1.69 ± 0.01 g of milk powder into the counterpoised scoop.

5.2 Addition of Acid to Butyrometer — Follow the procedure prescribed in **4.3.1** using the butyrometer (*see 2.1.2*).

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5.3 First Addition of Water — Add gently from the wash bottle sufficient cold water to form a layer about 6 mm deep on top of the acid, allowing the water to flow down the side of the bulb.

5.4 Addition of Sample — Insert the narrow end of the stemless funnel (*see* 2.2.3) into the neck of the butyrometer. Transfer the contents of the scoop to the funnel, removing the last particles with the camel hair brush. Tap the funnel gently until most of the powder is in the butyrometer. Transfer the powder remaining in the funnel to the butyrometer with the aid of the glass rod and the camel hair brush. Remove the funnel.

5.5 Addition of Amyl Alcohol — Measure one millilitre of amyl alcohol (*see* 3.2) into the butyrometer by means of the 1-ml pipette for amyl alcohol or the automatic measure (*see* 2.1.5). Do not wet the neck of the butyrometer with alcohol.

5.6 Second Addition of Water — Add hot water (70°C) from the wash bottle until the butyrometer is filled to about 5 mm below the shoulder, allowing all air entrained in the powder to escape.

5.7 Follow the procedures prescribed in 4.2.3 to 4.2.10. Read the butyrometer to the nearest 0.05 percent.

5.8 Third Centrifuging — Repeat the procedure described in 4.2.8 if the reading obtained after the second centrifuging is higher than that obtained after the first centrifuging.

5.9 Calculation of Percentage of Fat — Multiply the scale reading of the butyrometer by the factor 20/3 to obtain the percentage of fat in milk powders.

6. METHOD FOR CHEESE

6.1 Preparation of Sample — Grate samples of hard cheese. Grind samples of soft cheese. Mix thoroughly.

6.2 Weighing of Sample — Counterbalance the stoppered funnel (*see* 2.2.7) with its stopper inserted. Weigh 3 ± 0.01 g of the sample into the funnel.

6.3 Addition of Acid to Butyrometer — Follow the procedure prescribed in 4.3.1 using the cheese butyrometer (*see* 2.1.3).

6.4 First Addition of Water — Add gently from the wash bottle sufficient warm water (30 to 40°C) to form a layer about 6 mm deep on top of the acid, allowing the water to flow down the side of the bulb.

6.5 Addition of Sample — Insert the neck of the funnel containing 3 g of cheese into the neck of the butyrometer. Withdraw the stopper from the neck of the funnel and transfer all the cheese to the butyrometer with the aid of the glass rod or spatula.

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6.6 Addition of Amyl Alcohol — Follow the procedure prescribed in 5.5.

6.7 Second Addition of Water — Add warm water (30 to 40°C) from the wash bottle until the butyrometer is filled to about 5 mm below the shoulder.

6.8 Follow the procedures described in 4.2.3 to 4.2.10. Read the butyrometer to the nearest 0.3 percent, that is, one-third of the smallest scale division.

NOTE — For cheese containing more than 40 percent of fat, 1.5 g of cheese should be taken for the test and the butyrometer reading multiplied by 2.

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